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Spring 2025 ARML Power Contest

The Solutions

1. The first 10 terms of $F(3, 10)$ are 3, 10, 13, 23, 36, 59, 95, 154, 249, 403. $F(3, 10)_0 = 10 - 3 = 7$ and $F(3, 10)_{-1} = 3 - 7 = -4$.

2. Since $a \leq b \leq 5$, there are not very many possibilities. Since each sequence is non-decreasing, once it gets beyond 50 it can never return. Additionally, $F(2, 3) = 2$, 3, 5, ... is part of $F(1, 1)$ (and $F(1, 2)$) which already have smaller b's than $F(2, 3)$ so that sequence needn't be listed out either. So list all the sequences with $b \leq 5$ that are not part of previously listed sequences, until it grows beyond 50. Thus:

a	b	Sequence
1	1	$F(1, 1) = 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$
1	2	$F(1, 2)$ contained in $F(1, 1)$
2	2	$F(2, 2) = 2, 2, 4, 6, 10, 16, 26, 42, 68, \ldots$
1	3	$F(1, 3) = 1, 3, 4, 7, 11, 18, 29, 47, 76, \ldots$
2	3	$F(2, 3)$ contained in $F(1, 1)$
3	3	$F(3, 3) = 3, 3, 6, 9, 15, 24, 39, 63, \ldots$
1	4	$F(1, 4) = 1, 4, 5, 9, 14, 23, 37, 60, \ldots$
2	4	$F(2, 4) = 2, 4, 6, 10, 16, 26, 42, 68, \ldots$
3	4	$F(3, 4)$ contained in $F(1, 3)$
4	4	$F(4, 4) = 4, 4, 8, 12, 20, 32, 52, \ldots$
1	5	$F(1, 5) = 1, 5, 6, 11, 17, 28, 45, 73, \ldots$
2	5	$F(2, 5) = 2, 5, 7, 12, 19, 31, \boxed{50} \ldots$
3	5	$F(3, 5)$ contained in $F(1, 1)$
4	5	$F(4, 5) =$ contained in $F(1, 4)$
5	5	$F(5, 5) = 5, 5, 10, 15, 25, 40, 65, \ldots$

   There are no other Fibonacci-like sequences with second term $b \leq 5$, and the first (and only) one that showed up here was $F(2, 5)$.

3. Note that the number 22 does not show up in the table for the previous problem. Thus, $B(22) \geq 5$. Trying sequences with $b = 6$ will quickly result in $B(2, 6) = 2, 6, 8, 14, 22 \cdots$ so $B(22) = 6$.

4. This will be proven by induction. Note that it is true for $k = 3$ and $k = 4$. Now assume the relationship holds for any two consecutive values of the index, say $k - 2$ and $k - 1$. Then

$$\begin{aligned} F(a, b)_k &= F(a, b)_{k-2} + F(a, b)_{k-1} \\ &= (F_{k-4}a + F_{k-3}b) + (F_{k-3}a + Fk - 2b \\ &= (F_{k-4} + F_{k-3})a + (F_{k-3} + F_{k-2})b \\ &= F_{k-2}a + F_{k-1}b \end{aligned}$$

which is the desired relationship. (Note: you must demonstrate two base cases or use some form of strong induction for full credit!)

5. From the previous problem, $F(a,b)_k = F_{k-2}a + F_{k-1}b$. Similarly, $F(a',b)_k = F_{k-2}a' + F_{k-1}b$. Setting these equal, the b terms are identical and may be canceled. Then, since $k > 0$ there are no two Fibonacci numbers that are the same, so the F_{k-2} may be canceled as well, leaving the desired result.

6. Since $j \geq 1$, F_{k-j} is at most F_{k-1}. So $\dfrac{F_k - F_{k-j}}{F_{k-j-1}} \geq \dfrac{F_k - F_{k-1}}{F_{k-j-1}} = \dfrac{F_{k-2}}{F_{k-j-1}}$. Then, since $j \geq 0$, $F_{k-j-1} \leq F_{k-2}$. Thus $\dfrac{F_k - F_{k-j}}{F_{k-j-1}} \geq \dfrac{F_{k-2}}{F_{k-2}} = 1$. Equality cannot hold unless $j = 1$ as if $j > 1$ all the foregoing inequalities are strict. When $j = 1$ the fraction is exactly equal to 1.

7. Assume $b' \leq b$. Now from problem 4 $F(a,b)_k = F_{k-2}a + F_{k-1}b$ and $F(a',b')_{k-j} = F_{k-j-2}a' + F_{k-j-1}b'$. Setting these equal and noting that all numbers are positive,

$$\begin{aligned} F_{k-j-2}a' + F_{k-j-1}b' = F_{k-2}a + F_{k-1}b &> F_{k-1}b \\ F_{k-j-2}a' &> F_{k-1}b - F_{k-j-1}b' \geq (F_{k-1} - F_{k-j-1})b' \\ a' &> \frac{F_{k-1} - F_{k-j-1}}{F_{k-j-2}}b'. \end{aligned}$$

   By the previous problem, the fraction is at least 1, thus $a' > b'$, contradicting the nondecreasing requirement for the sequence to be Fibonacci-like.

8. By assumption, $b_{k-1} = a_{k-1} - 1 = a_{k-1} + (-1)^1 F_1$. Since $b_{k-2} + b_{k-1} = n = a_{k-2} + a_{k-1}$ and $b_{k-1} = a_{k-1} - 1$, subtracting $a_{k-1} - 1$ from both sides of this equation yields $b_{k-2} = a_{k-2} + 1 = a_{k-2} + (-1)^2 F_2$. So the desired result is true for $j = 1$ and $j = 2$. Now assume the result is true for $j = r - 2$ and $j = r - 1$ for some $r \geq 3$. Then

$$\begin{aligned} b_{k-r} &= b_{k-(r-2)} - b_{k-(r-1)} \\ &= a_{k-(r-2)} + (-1)^{r-2}F_{r-2} - a_{k-(r-1)} - (-1)^{r-1}F_{r-1} \\ &= (a_{k-(r-2)} - a_{k-(r-1)}) + (-1)^{r-2}(F_{r-2} + F_{r-1}) \\ &= a_{k-r} + (-1)^r F_r \end{aligned}$$

   and the proof is complete by induction.

9. (a) $17 - 11 = 6$, $11 - 6 = 5$, $6 - 5 = 1$, $5 - 1 = 4$, but $4 > 1$, so the sequence extends to $1, 5, 6, 11, 17, \ldots$.

   (b) $8 - 6 = 2$, $6 - 2 = 4$, but $4 > 2$, so the sequence extends only one term to $2, 6, 8, \ldots$.

   (c) $16 - 7 = 9$ but already $9 > 7$ so this sequence cannot be extended backward and remain Fibonacci-like.

10. To extend the sequence backward, the previous term would be $b - a$. So the sequence cannot be extended backward unless $0 < b - a \leq a$. Since $a \leq b$, the inequality $0 < b - a$ is true unless $a = b$. Requiring $b - a \leq a$ is the same as $b \leq 2a$. So the sequence can be extended backward only if $a < b \leq 2a$.

(a) Assume the sequence can be extended backward at least one term. Using the Fibonacci recursion again, the next prior term would be $a-(b-a)=2a-b$. The sequence can be extended backward at least two terms when $0<2a-b\le b-a$. So extension by exactly one term requires $a<b\le 2a$ and *either* ($2a-b\le 0$ or $2a-b>b-a$). The first of the conditions in the *or* statement is $2a-b\le 0$ which is only compatible with $a<b\le 2a$ when $b=2a$. The second is compatible whenever $\frac{2}{3}b<a<b$.

Basically, when $2a=b$ the sequence $a, 2a, \ldots$ can be extended to $a, a, 2a, \ldots$ but the next term going backward would be zero, which is not allowed. When $a>2b/3$ the sequence can be extended backward by one term, but that term will be less than $a/2$ precluding going back another term.

(b) Combining the requirements found above, the sequence can be extended backward at least two terms when $b/2<a\le 2b/3$.

11. Applying the technique to $n=199$:

Start:							1	198	199
Alter:							2	197	199
Alter:							⋮	⋮	⋮
Alter:							67	132	199
Lengthen:						2	65	67	132	199
Alter:						5	63	68	131	199
Alter:						⋮	⋮	⋮	⋮	⋮
Alter:						26	49	75	124	199
Lengthen:				3	23	26	49	75	124	199
Alter:				11	18	29	47	76	123	199
Lengthen:	1	$\boxed{3}$	4	7	11	18	29	47	76	123	199

This can't be lengthened, and the next alteration will create a negative term, so the process stops here, with $B(199)=3$. Next, for $n=201$:

Start:						1	200	201
Alter:						2	199	201
Alter:						⋮	⋮	⋮
Alter:						67	134	201
Lengthen:					67	67	134	201
Alter:					65	68	133	201
Lengthen:				3	65	68	133	201
Alter:				⋮	⋮	⋮	⋮	⋮
Alter:				27	49	76	125	201
Lengthen:		5	22	27	49	76	125	201
Alter:		13	17	30	47	77	124	201
Lengthen:	4	$\boxed{13}$	17	30	47	77	124	201

Again, this cannot be lengthened, and the next alteration will create a negative term, to the process terminates with $B(201) = 13$.

12. First, show the process must come to a stop after a finite number of steps. The process begins with the sequence $1, n-1, n$. Each time the alteration step is executed, the penultimate term decreases by 1 and the next previous term increases by 1. Thus, after the first time the alteration step is executed, the sequence is $\ldots, 2, n-2, n$ and then $\ldots, 3, n-3, n$. But this certainly cannot occur more than $\lfloor n/2 \rfloor$ times, or these terms will no longer be nondecreasing.

    The sequence cannot be lengthened to more than $n+1$ terms (in fact, the sequence can only be $n+1$ terms long if $n = 1$, 2, or 3). This is because $a_{k+2} = a_k + a_{k+1}$ by the Fibonacci recurrence, and since the sequence is nondecreasing, $a_k + a_{k+1} \leq 2a_k$. Thus if $n \leq 2^s$ then no Fibonacci-like sequence containing n can have more than $2s+1$ terms before reaching n. (This can be formalized by induction, if desired.) Thus there are a finite number of lengthening steps and a finite number of alteration steps, so the algorithm terminates after a finite number of steps.

    Now to show that when it does terminate, the longest Fibonacci-like sequence containing n has been found. So assume the algorithm terminates with a sequence whose final three terms are $k, n-k, n$. Certainly no sequence that ends $j, n-j, n$ with $j < k$ is as long or longer than the sequence under consideration, for the alteration process already considered that sequence, and it could not be lengthened at the time it was considered. Is it possible that there could be a longer sequence with $j > k$?

    Let the sequence that the algorithm finds be $A_1, A_2, \ldots, A_{r-2} = k, A_{r-1} = n-k, A_r = n$. This cannot be altered by decreasing the penultimate term. The obstruction to doing so must be that the resulting sequence either has a non-positive term or decreases from one term to the next. That is, either $A_{r-s} - F_s < 0$ because the alteration made this term decrease via the formula in problem 8, or there is a pair of consecutive terms $A_{r-s} \leq A_{r-s+1}$ but the altered terms become $A_{r-s} + F_s > A_{r-s+1} - F_{s-1}$, again using the formula from problem 8. But in the first case, altering the sequence again continues to decrease A_{r-s} by repeatedly subtracting F_s, so no sequence with $j > k$ could have a positive number in position $r-s$. Similarly, the failure to be nondecreasing continues to be exacerbated the more the sequence is altered.

    Thus, once the algorithm encounters a sequence that cannot be either lengthened or altered by decreasing the penultimate term, so sequence obtained by further alteration can remain Fibonacci-like and have as many terms before reaching n. Thus, the longest sequence leading up to n has been found and the proof of the algorithm's correctness is complete.

13. This is a simple consequence of the conditions discovered in problem 10. The sequence can be lengthened if and only if $a < b \leq 2a$. Thus it cannot be lengthened if $b \leq a$ or if $b > 2a$. Since $a \leq b$, the first of those two conditions would require $a = b$ while the second is stronger than $a \leq b$. So b can only occur as the second term of the sequence if $2a < b$ or if $a = b$. There are $\lfloor (b-1)/2 \rfloor$ positive integers a that satisfy $2a < b$ and one more for $a = b$. For example, none of the sequences

$F(1,6)$, $F(2,6)$, $F(6,6)$ can be lengthened, while $F(3,6)$ can be lengthened to $F(3,3)$, $F(4,6)$ can be lengthened to $F(2,2)$, and $F(5,6)$ to $F(1,5)$. Note that $b = 6$ and $1 + \lfloor (b-1)/2 \rfloor = 1 + \lfloor 5/2 \rfloor = 1 + 2 = 3$.

14. If $B(n) = 1$, then n must be a member of a sequence of the form $F(a,1)$. Since the sequence must be nondecreasing, $a = 1$. Now $F(1,1)$ is the standard Fibonacci sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ... So each of those numbers have $B(n) = 1$, and the smallest such is $L(1) = 1$.

    Now $b = 2$ can only occur in $F(2,2) = 2$, 2, 4, 6, 10, 16, 26, 42, 68,.... The smallest number we haven't seen yet is $n = 4$. $B(4) = 2$ and 4 is the smallest number with $B(4) = 2$, so $L(2) = 4$.

    Combining these lists, numbers that are no longer candidates to be $L(n)$ are 1, 2, 3, 4, 5, 6, 8, 10, 13, 16, 21, 26, 34, 42, 55, 68, 89.

    Next, $b = 3$ can occur in $F(1,3) = 1$, 3, 4, 7, 11, 18, 29, 47, 76, ...and $F(3,3) = 3$, 3, 6, 9, 15, 24, 39, 63, ... The smallest number in either of these lists that hasn't been seen for a smaller b is $n = 7$, so $L(3) = 7$. Our list of non-candidates grows to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 15, 16, 18, 21, 24, 26, 29, 34, 39, 42, 47, 55, 63, 68, 89.

    For $b = 4$, there is $F(1,4) = 1$, 4, 5, 9, 14, 23, 37, 60, 97, ...and $F(4,4) = 4$, 4, 8, 12, 20, 32, 52, 84, ... The smallest number that is a candidate is 12, so $L(4) = 12$. Non-candidates are now 1–16, 18, 20, 21, 23, 24, 26, 29, 32, 34, 37, 39, 42, 47, 52, 55, 60, 63, 68, 84, 89, 97.

    For $b = 5$, there are three possibilities: $F(1,5) = 1$, 5, 6, 11, 17, 28, 45, 73, ..., $F(2,5) = 2$, 5, 7, 12, 19, 31, 50, 81, ..., and $F(5,5) = 5$, 5, 10, 15, 25, 40, 65, ... So $L(5) = 17$. The non-candidates list grows to 1–21, 23–26, 28, 29, 31, 32, 34, 37, 39, 40, 42, 45, 47, 50, 52, 55, 60, 63, 65, 68, 73, 84, 89, 97.

    Next comes $b = 6$. $F(1,6) = 1$, 6, 7, 13, 20, 33, 53, 86, ..., $F(2,6) = 2$, 6, 8, 14, 22, 36, 58, 94, ..., and $F(6,6) = 6$, 6, 12, 18, 30, 48, 78, ... So $L(6) = 22$. There are now fewer candidates less than 100 than non-candidates. Those are 27, 35, 38, 41, 43, 44, 46, 49, 51, 54, 56, 57, 59, 61, 62, 64, 66, 67, 69, 70, 71, 72, 74, 75, 76, 77, 79–83, 85, 87, 88, 90–96, 98, 99.

    When $b = 7$ the sequences are $F(1,7) = 1$, 7, 8, 15, 23, 38, 61, 99..., $F(2,7) = 2$, 7, 9, 16, 25, 41, 66..., $F(3,7) = 3$, 7, 10, 17, 27, 44, 71, ..., and $F(7,7) = 7$, 7, 14, 21, 35, 56, 91, ... So $L(7) = 27$, and the candidates list reduces to 43, 46, 49, 51, 54, 57, 59, 62, 64, 66, 67, 69, 70, 72, 74, 75, 76, 77, 79–83, 85, 87, 88, 90, 92–96, 98.

    Continuing, $b = 8$ has $F(1,8) = 1$, 8, 9, 17, 26, 43, 69, ..., $F(2,8) = 2$, 8, 10, 18, 28, 46, 74..., $F(3,8) = 3$, 8, 11, 19, 30, 49, 79, ..., and $F(8,8) = 8$, 8, 16, 24, 40, 64, ... This leaves $L(8) = 43$. Remaining candidates: 51, 54, 57, 59, 62, 66, 67, 70, 72, 75, 76, 77, 80–83, 85, 87, 88, 90, 92–96, 98.

    Continuing this process (it would have been easier to have a table of numbers from 1–100 which were the candidates and strike them out as they occurred in the various sequences, so the list didn't have to be repeated, which is not easily done in printed form!) finds $L(9) = 51$ and $L(10) = 59$.

So the results are

b	1	2	3	4	5	6	7	8	9	10
$L(b)$	1	4	7	12	17	22	27	43	51	59

You might note that this table has lots more Fibonacci numbers hidden in it. For instance, the values of L have jumps of 3, then 5, then (mostly) 8. If you extend the table, the runs of a particular jump size have lengths 2, 4, 6, 10, 16, ... which are doubles of Fibonacci numbers. These can be explained (with a bit of computation) are the article pointed out in the original question has all the answers. Enjoy!